Sprint Report 3

Unit code: EAT40005
Unit Name: Engineering Project A
Submission date: 07/06/2026

Student Name	Student Id	Statement of contribution to the report
Rumeth Gunasinghe	104498227	Developed the Sprint Plan, Sprint Progress and Sprint Demonstration sections. Contributed to MuJoCo simulation development, humanoid balancing implementation, controller tuning, technical documentation and demonstration preparation.
Thinidu Manapperuma	104675729	Developed the Sprint Review, Retrospect, Lessons Learned and Future Plans sections. Contributed to project evaluation, critical analysis, sprint reflection, documentation review and future project planning.

ACKNOWLEDGMENT OF COUNTRY

We acknowledge the Traditional Owners of the lands on which this work was completed. We recognise the Wurundjeri Woi Wurrung people of the Kulin Nation as the Traditional Custodians of the land in Melbourne, Victoria, Australia, where this project work was undertaken.

We pay our respects to their Elders past, present and emerging, and acknowledge their continuing connection to land, waters, culture and community.

As students living and studying in Australia, we recognise the important role of Indigenous knowledge and perspectives and their contribution to sustainable and responsible technological development.

1. SPRINT PLAN

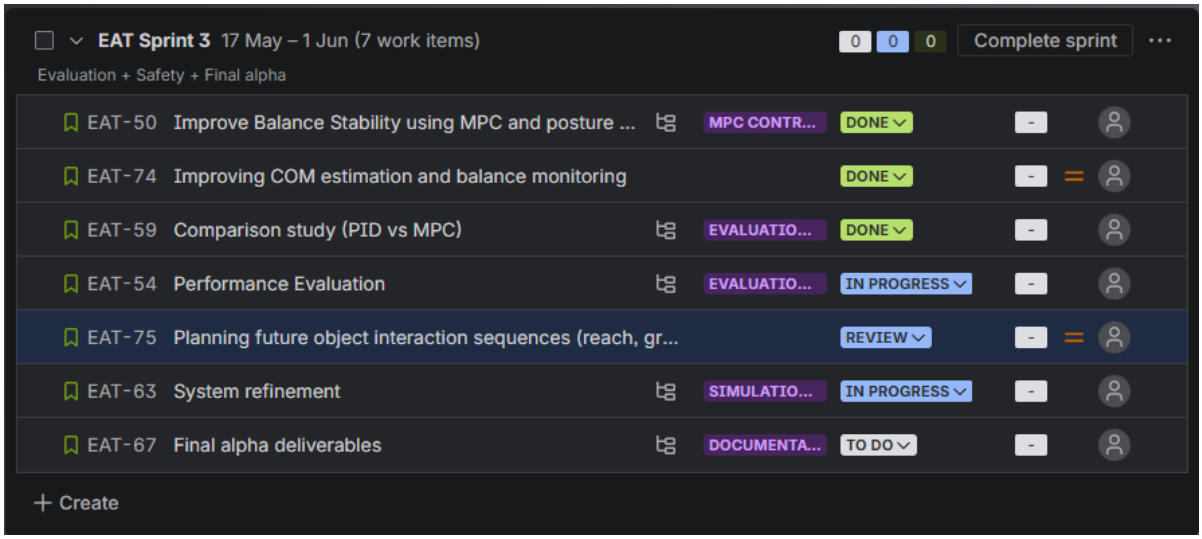


Figure 1: Sprint Backlog

Sprint 3 focused on moving the project from static humanoid balancing to autonomous locomotion and environment interaction. During Sprint 2, the MuJoCo-based humanoid balancing framework was successfully developed using the Unitree H1 robot model, incorporating PID stabilization, MPC-inspired balance prediction and center-of-mass (COM) estimation. While the robot was capable of maintaining temporary balance, locomotion and object interaction capabilities weren't implemented.

The primary objective of Sprint 3 was to extend the balancing framework into a foundation for humanoid navigation. The team planned to investigate how Model Predictive Control (MPC) could be used not only for balance stabilization but also for generating movement trajectories toward a target position. The ultimate goal was for the humanoid robot to identify a target location, walk toward it while maintaining stability and eventually perform basic interaction tasks.

The sprint backlog included:

- Improving balance stability using enhanced MPC and posture control
- Investigating recovery stepping behaviour for balance preservation
- Designing locomotion state transitions for walking behaviour
- Improving COM estimation and balance monitoring
- Researching humanoid gait generation techniques
- Investigating ROS2 and Isaac Sim humanoid control pipelines
- Designing target navigation architecture
- Planning future object interaction sequences (reach, grasp and lift)

The expected outcome of the sprint was a humanoid control framework capable of progressing from static balance toward controlled locomotion while maintaining overall system stability

Some of the outcomes werem't achieved before the due date due to complexity of the tasks and the lack of technical knowledge.

2. SPRINT PROGRESS

The first major activity involved continued refinement of the balancing controller. Extensive tuning was performed on hip, knee and ankle controllers to improve stability duration and reduce oscillatory behaviour. Multiple controller configurations were tested to determine how posture geometry affected overall balance performance.

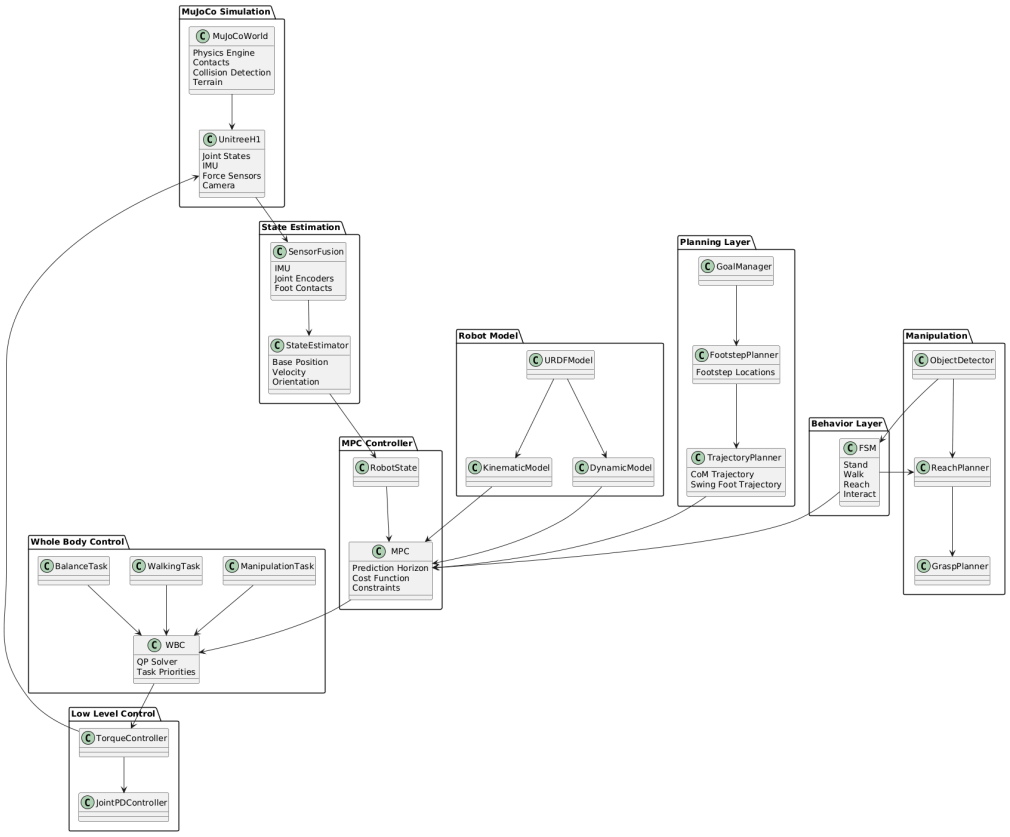


Figure 2: The system architecture of the FInal prototype

A major finding during the sprint was that humanoid balance depended heavily on posture geometry rather than controller gains alone. Testing revealed that straight-leg standing configurations resulted in immediate instability, while crouched postures with bent knees significantly improved center-of-mass alignment and stability duration. As a result, a nominal crouched standing pose was developed and integrated into the controller.

The team further investigated center-of-mass behaviour during balancing. A simplified COM estimation module was refined to monitor body position and velocity during simulation. Through repeated testing, COM drift was identified as one of the primary causes of balance

failure. Although the controller could temporarily stabilize the robot, accumulated COM drift eventually caused the robot to move outside its support region.

The MPC-inspired balancing module developed in Sprint 2 was further evaluated. Future pitch prediction and balance estimation techniques were tested as potential foundations for future locomotion planning. While full MPC locomotion was not achieved during the sprint, the work provided valuable understanding of how predictive control could be integrated into future walking systems.

Additional investigation was conducted into the kinematic structure of the Unitree H1 humanoid. Joint mappings, actuator behaviour and pelvis-torso interactions were analysed. During this process, torso and pelvis alignment issues were identified as significant contributors to instability. Various stabilization strategies were explored to reduce rotational drift and improve body alignment.

Research activities were also conducted into modern humanoid robotics workflows. NVIDIA Isaac Sim, ROS2 integration, IMU-based state estimation and reinforcement learning locomotion pipelines were investigated to better understand industry-standard approaches for humanoid control.

At the conclusion of the sprint, the robot demonstrated a semi-stable crouched standing posture lasting approximately 4–5 seconds before instability accumulated. While locomotion and object interaction were not achieved, the sprint successfully identified the primary technical challenges that must be addressed before reliable walking behaviour can be implemented.

3. SPRINT DEMONSTRATION

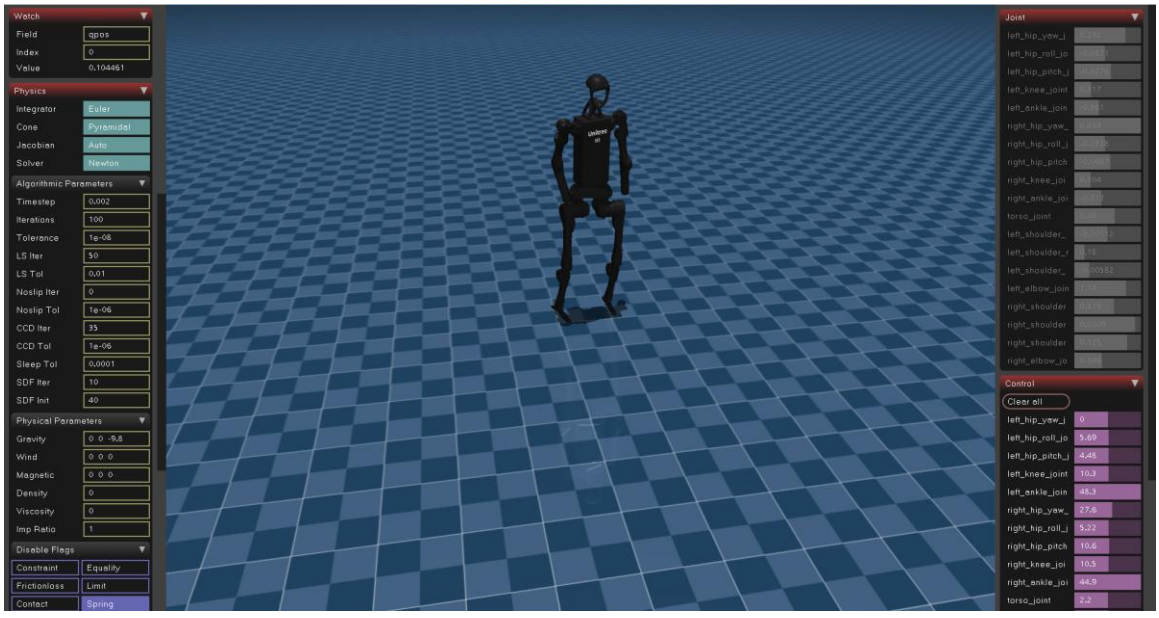


Figure 3 : Used parameters in the Environment

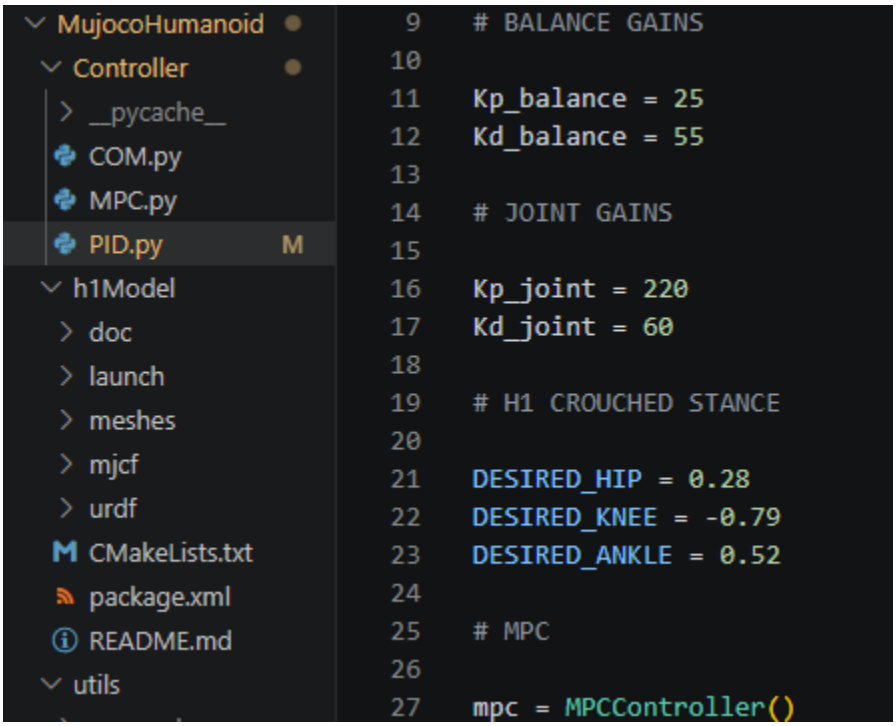


Figure 4: Desired angles for a balanced standing state

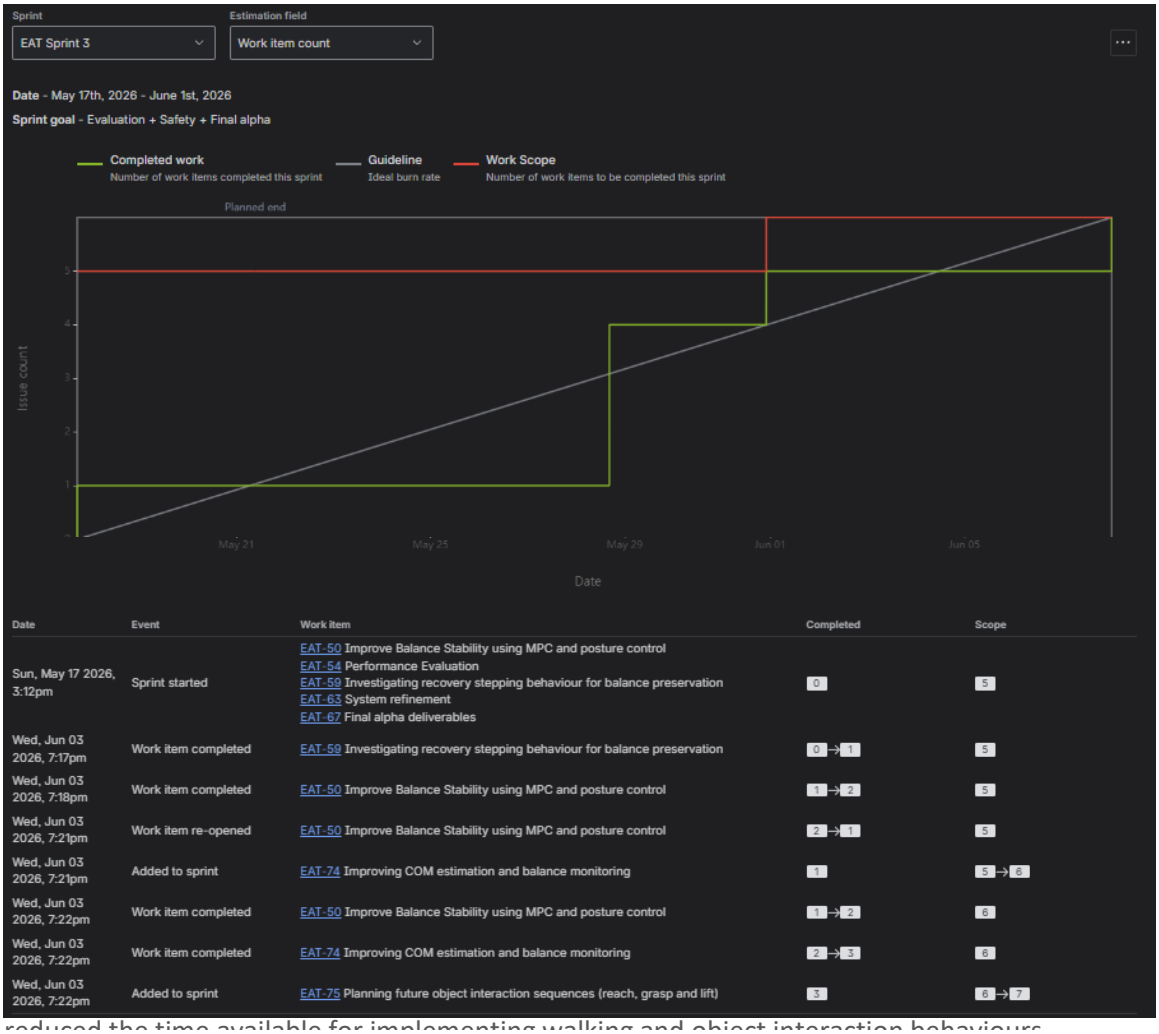
The balancing behaviour of the robot was demonstrated in real time. The robot successfully entered a crouched standing posture and maintained temporary balance through active stabilization. The demonstration highlighted the improvements achieved since earlier project stages, where the robot collapsed immediately upon simulation start.

The team discussed the posture optimization process, including the effect of knee flexion, ankle angles and torso alignment on overall stability. Controller tuning experiments and the impact of COM positioning were also presented.

The client and supervisor were informed that locomotion and object interaction had not yet been achieved. Instead, the focus of the sprint had shifted toward understanding and resolving the stability limitations preventing reliable walking behaviour.

- The balancing framework represented meaningful progress from previous demonstrations.
- Greater emphasis should be placed on achieving reliable balance before introducing locomotion.
- Future work should focus on improving stability duration and implementing recovery stepping behaviour.
- The team should continue investigating modern humanoid robotics workflows such as ROS2, Isaac Sim and reinforcement learning locomotion approaches.

4. SPRINT REVIEW (CRITICAL REVIEW OF THE PRODUCT)



Date	Event	Work item	Completed	Scope
Sun, May 17 2026, 3:12pm	Sprint started	EAT-50 Improve Balance Stability using MPC and posture control EAT-54 Performance Evaluation EAT-59 Investigating recovery stepping behaviour for balance preservation EAT-63 System refinement EAT-67 Final alpha deliverables	0	5
Wed, Jun 03 2026, 7:17pm	Work item completed	EAT-59 Investigating recovery stepping behaviour for balance preservation	0→1	5
Wed, Jun 03 2026, 7:18pm	Work item completed	EAT-50 Improve Balance Stability using MPC and posture control	1→2	5
Wed, Jun 03 2026, 7:21pm	Work item re-opened	EAT-50 Improve Balance Stability using MPC and posture control	2→1	5
Wed, Jun 03 2026, 7:21pm	Added to sprint	EAT-74 Improving COM estimation and balance monitoring	1	5→6
Wed, Jun 03 2026, 7:22pm	Work item completed	EAT-50 Improve Balance Stability using MPC and posture control	1→2	6
Wed, Jun 03 2026, 7:22pm	Work item completed	EAT-74 Improving COM estimation and balance monitoring	2→3	6
Wed, Jun 03 2026, 7:22pm	Added to sprint	EAT-75 Planning future object interaction sequences (reach, grasp and lift)	3	6→7

reduced the time available for implementing walking and object interaction behaviours

Figure 5: Sprint Burndown chart

During the sprint demonstration, the client acknowledged the technical progress made in developing the balancing framework but noted that the project was behind the originally expected schedule. The client expected greater progress toward locomotion and autonomous task execution by the end of Sprint 3. Feedback indicated that future work should place greater emphasis on achieving practical locomotion milestones while continuing to improve balance stability.

The team agreed with the client's assessment that progress toward the final project objectives had been slower than planned. However, the demonstration also highlighted that the primary limitations were not due to a lack of effort but rather the complexity of humanoid robotics and the steep learning curve associated with advanced simulation and control systems. The balancing framework developed during Sprint 3 provides a stronger technical foundation for future work, even though the planned locomotion objectives were not fully achieved.

Going forward, the team intends to improve task estimation accuracy, break large technical challenges into smaller milestones, and allocate more dedicated time to implementation after research activities are completed. Future development will focus on recovery stepping, locomotion state machines, target navigation and object interaction behaviours while building upon the balancing framework established during this sprint.

5. RETROSPECT (CRITICAL REVIEW OF THE PROCESS)

sDuring Sprint 3, reasonable time was spent investigating the relationship between the humanoid robot's center of mass and standing stability. Lot of standing postures were tested in the MuJoCo simulation environment to measure how torso inclination, hip angles and knee flexion will affect the balance performance. multiple unstable configurations resulted in the robot falling forward or backward soon after initialization, requiring iterative adjustments to posture parameters.

A huge portion of the development effort was put into controller tuning and debugging. Joint level PID gains were back and fourth modified and tested to minimise oscillations when maintaining sufficient corrective torque. It was seen that increasing controller gains improved responsiveness but mostly introduced instability due to overshooting behaviour. Looking for a suitable balance between responsiveness and stability required extensive trial and error experimentation.

The implementation of the MPC framework also required additional debugging effort. first prediction models produced unrealistic control outputs because of not so accurate state information and simplified assumptions for robot dynamics. To address this issue, state variables including base position, body orientation and center of mass location were again and again monitored and validated throughout the simulation process.

Frequent simulation testing was performed after each of the controller modification. This repeating approach allowed rapid identification of system weaknesses and making incremental improvements to standing performance. At the end of the sprint, the robot demonstrated noticeably improved balance duration and posture stability compared to the initial implementation.

6. LESSONS LEARNED (CRITICAL REVIEW OF SPRINT ONE EXPERIENCE AND FUTURE PLAN)

aOne major lesson learned was that successful humanoid balancing requires correct modelling of the robot's physical structure. minor errors in joint configuration, body dimensions or mass distribution majorly affected controller performance and mostly resulted in unstable behaviour. This shows the importance of making sure the robot model before focusing on advanced control strategies.

One more lesson learned was the value of simulation driven development. MuJoCo provided instant feedback regarding controller modifications and enabled fast experimentation without risking physical hardware damage. Visualization of joint behaviour, contact forces and body posture helps identify problems that were hard to detect through number output alone.

The sprint also showed that implementing MPC alone is insufficient for achieving reliable humanoid locomotion. Supporting systems such as state estimation, foot contact detection and trajectory planning are similarly important components of a complete locomotion framework. Future development must have focus on making these subsystems other than relying completely on controller improvements.

Future work will build on the recent balancing framework by introducing dynamic walking behaviours. Planned developments will have footstep generation, swing foot trajectory planning, disturbance recovery mechanisms and target directed navigation. Additional developments will also focus on refining center of mass estimation and placing perception systems to support future object interaction tasks.

The knowledge gained throughout Sprint 3 has given us a stronger understanding of humanoid balance control, controller tuning methodologies and simulation based robotics development, providing a solid foundation for the remaining stages of the project.

7. FUTURE PLANS

Based on the outcomes of Sprint 3, future development will focus on improving the current balancing framework and progressing toward autonomous locomotion and object interaction.

Recovery Stepping Behaviour

The current robot attempts to balance in place but cannot recover once instability becomes too large. Future work will focus on implementing recovery stepping so the robot can reposition its feet and regain balance rather than falling.

Locomotion State Machine

A locomotion state machine will be developed to manage different movement behaviours such as standing, walking, turning and stopping. This will provide a structured framework for controlling humanoid movement.

Advanced MPC Development

The current MPC-inspired controller is primarily used for balance prediction. Future work will extend MPC capabilities to generate walking trajectories and predictive movement plans while maintaining stability constraints.

Improved COM Estimation

More accurate center-of-mass estimation and state prediction techniques will be investigated to improve balance awareness and allow the controller to make better stabilization decisions.

Target Navigation

The robot will be enhanced with target navigation capabilities, allowing it to identify a destination and generate movement commands to reach that location autonomously.

Object Interaction

Following successful locomotion, object interaction behaviours such as reaching, grasping and lifting will be implemented. This will allow the humanoid to perform simple task-oriented actions within the environment.

Reinforcement Learning Integration

The team plans to investigate reinforcement learning approaches to complement traditional control methods. This may allow the robot to learn more robust balancing and locomotion strategies through training.

ROS2 and Isaac Sim Integration

Industry-standard robotics frameworks such as ROS2 and NVIDIA Isaac Sim will be explored to improve scalability, enable additional sensor integration and support future development.

Improved Testing and Evaluation

More structured testing procedures, parameter logging and performance evaluation methods will be introduced to improve experimentation, debugging and controller tuning efficiency.

The long-term objective remains the development of a humanoid robot capable of balancing, navigating toward a target and interacting with objects autonomously within a simulated environment.